

Anirudh Malik

Data Scientist — AI/ML Engineer

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LinkedIn GitHub Portfolio Gurugram, India

SUMMARY

Data Scientist and AI/ML Engineer with strong foundations in **predictive modeling, recommender systems, text mining, statistical modeling, and hypothesis testing**. Experienced in building **advanced data analytics pipelines, scalable algorithms, and high-dimensional modeling frameworks**. Skilled at delivering **production-grade ML solutions** across cloud platforms (AWS, Azure), deploying containerized systems with **Docker, Kubernetes, Airflow, and MLflow**. Strong research background in **physics-inspired ML**, complemented with practical expertise in automation, data mining, and business-focused AI solutions.

EXPERIENCE

AI - Software Engineer *Dextra Labs*
Gurgaon, India Aug 2025 – Present

- Led the design of an **Excel-to-template automation system** that standardizes financial spreadsheets by extracting metadata (headers, sheet names, samples) and applying structured AI prompts with template validation. This improved accuracy, ensured consistent formatting, and created audit/debug logs for transparency.
- Delivered **SFS application automation** for video content and engagement: splitting session videos, embedding pre-rolls, adding SFS logos/watermarks, and automating posting directly to the platform.
- Automated **email notifications and segment updates** with OneSignal, streamlined RSVP onboarding with APIs, and integrated Google Sheets for registration error-tracking and triggers.
- Created reusable **HTML templates** for deeplink sharing and engagement campaigns. These automations reduced manual effort by 70%, scaled content delivery, and improved user onboarding and retention.

Research Associate *IIT Delhi*
Delhi, India 04/2025 – Present

- Developing a **hybrid anomaly detection framework** by integrating **VAEs** with **FROCC** for unsupervised jet tagging in high-dimensional particle physics data.
- Simulating **Higgs boson production** via $gg \rightarrow H \rightarrow \tau^+\tau^-$ using *MadGraph5*, *Pythia8*, and *Delphes*, generating datasets for analysis and model validation.
- Performing **exploratory data analysis (EDA)** on CMS MiniAOD datasets, comparing generator-level and reconstructed τ leptons to optimize data quality and selection.
- Designing **latent-space anomaly scores** and building **generative modelling workflows** to enable model-independent searches for beyond-the-Standard-Model physics.

AI/ML Specialist
Noida

Imagenators Global
05/2025 – 07/2025

- Built a **conversational assistant** using Gemini's voice models with real-time multilingual support, low latency, and interruption handling.
- Developed a containerized **RAG pipeline** (FastAPI, LangChain, FAISS, Gemini 1.5) to support PDF ingestion, data wrangling, and real-time LLM-based question answering.
- Benchmarked **generative AI vs classical NLP** for retrieval and social listening tasks, improving accuracy of domain-specific queries.

Trainee Developer
New Delhi

Dev Group
02/2025 – 05/2025

- Developed a **CO₂ emissions prediction tool** with regression techniques (Flask + MLflow). Conducted EDA, built dashboards, and created internal APIs for analytics and reporting.

Graduate Researcher

Univ. of Sheffield, UK
09/2023 – 09/2024

- Built **PICNN classifiers** for $Z \rightarrow e^+e^-$ classification using ATLAS MC data. Applied **Optimal Transport** to align AF3 and Geant4 simulations.

EDUCATION

- MSc in Particle Physics**, University of Sheffield, UK
2023 – 2024 First Division
- BSc in Physical Science**, University of Delhi, India
2020 – 2023 Distinction

SKILLS

- Programming:** Python, C++, SQL, Bash, Java
- ML/AI:** Predictive Modeling, Recommender Systems, Generative AI (VAEs, GANs, Transformers), Deep Learning (PyTorch, TensorFlow), NLP, Text Mining, Social Listening, Keras, Hugging Face, Pyspark
- Data Science:** Statistical Modeling, Hypothesis Testing, EDA, Feature Engineering, Anomaly Detection
- Big Data & MLOps:** Hadoop Ecosystem, Apache Spark, Docker, Kubernetes, Airflow, MLflow, CI/CD
- Frameworks/APIs:** FastAPI, Flask, Django, LangChain, Ollama, Mistral
- Data Eng. and Vis.:** Pandas, NumPy, PostgreSQL, MongoDB, SciKit Learn, Pandas, Matplotlib
- Cloud:** AWS (EC2, S3, Lambda, SageMaker), Azure
- Research Tools:** ROOT, Geant4, DDSCAT

CERTIFICATIONS

- Data Science, ML, DL, NLP
- Mastering SQL & Analytics

LANGUAGES

English — Professional

Hindi — Native